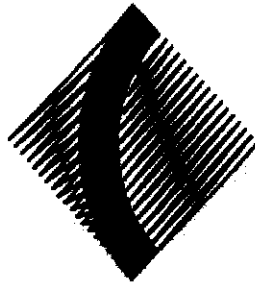


AW
AT
JG

Name: _____
Class: 12MTX_____
Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2009 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 1

*Time allowed - 2 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 7.

*****Each page must show your name and your class. *****

Question 1 (12 Marks)**Marks**

- (a) P (− 2, 3) and Q (6, −1) are two points in the number plane.

Find the coordinates of the point R that divides the interval PQ internally in the ratio 3:2.

2

- (b) Find the limiting sum of the geometric series

$$\left(\frac{e}{e+1}\right) + \left(\frac{e}{e+1}\right)^2 + \left(\frac{e}{e+1}\right)^3 + \dots$$

2

- (c) Solve the inequality $\frac{4}{1-x} \leq 3$ and graph your solution on a number line

3

- (d) The equation $x^3 + 2x^2 + 3x + 6 = 0$ has α , β and γ as its roots.

Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

2

- (e) Find $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

1

- (f) Find the acute angle between the lines: $x - \sqrt{3}y + 1 = 0$

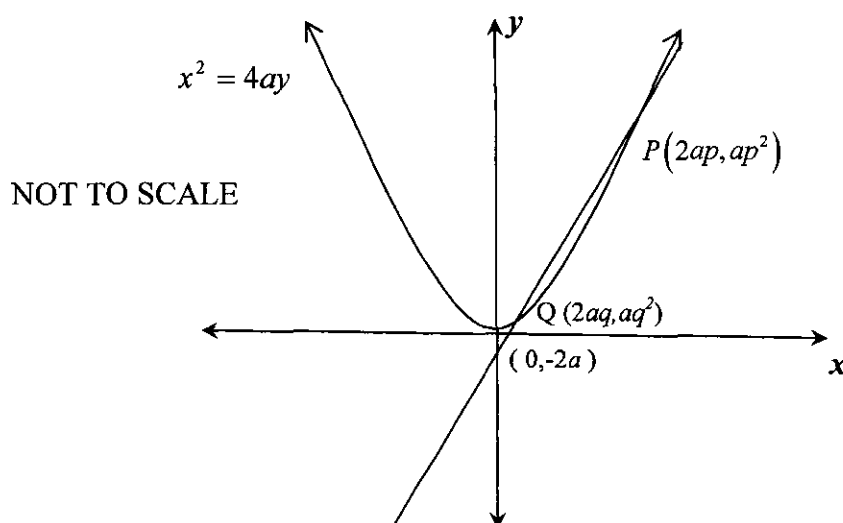
2

and $y = x - 4$.

Give your answer correct to the nearest degree.

Question 2 (12 Marks) START A NEW PAGE**Marks**

- (a) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$



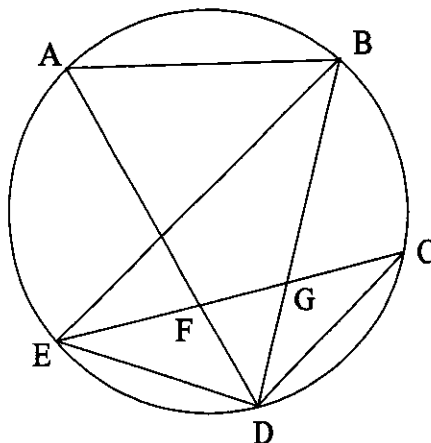
- (i) Given the gradient of the chord PQ is $\frac{p+q}{2}$, show that the equation of PQ is $2y = (p+q)x - 2apq$. 1
- (ii) The point joining P and Q passes through the point $(0, -2a)$. Show that $pq = 2$. 1
- (iii) The normals to the parabola $x^2 = 4ay$ at points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ intersect at K . The coordinates of K are $(-apq(p+q), a(p^2 + q^2 + pq + 2))$. **Do not prove this.**
- Prove that the locus of K is the parabola $x^2 = 4ay$. 2

Question 2 continues on the page 3.....

Question 2 continued.....

Marks

- (b) A, B, C, D and E are points on a circle such that $\angle DEC = \angle ECD$.



- (i) Give a reason why $\angle CED = \angle EBD$. 1
- (ii) Show that ABGF is a cyclic quadrilateral. 3
- (c) A tower CX is observed at an angle of elevation of 14° from a point A on level ground. The same tower is observed from B, 1 km from A, with an angle of elevation of 17° . $\angle ACB = 120^\circ$. C is the base of the tower.
- (i) Draw a diagram showing this information. 1
- (ii) Calculate h , the height of the tower CX. Give your answer correct to the nearest metre. 3

Question 3 (12 Marks) START A NEW PAGE**Marks**

- (a) Consider graph of the function $h(x) = \frac{3x}{1-x^2}$
- (i) Find the equation of any asymptotes. **2**
- (ii) Show why $h(x)$ has no turning points. **2**
- (iii) Sketch $h(x)$ showing any asymptotes and intercepts. **2**
- (b) A polynomial $P(x)$ of degree three, has zeros at $x = 1$, $x = -1$ and $x = 2$, and a remainder of 16 when divided by $(x - 3)$.
Find $P(x)$, expressing it in the form $P(x) = P_0x^3 + P_1x^2 + P_2x + P_3$ **2**
- (c) The area bounded by the x axis and the part of the curve $y = \sin x$ from $x = 0$ to $x = \pi$ is rotated about the x axis to form a solid.
Find the exact volume of the solid. **2**
- (d) Using the substitution $u = x^4$, find $\int \frac{x^3}{1+x^8} dx$. **2**

Question 4 (12 Marks) START A NEW PAGE

- (a) Consider the function $f(x) = (x + 2)^2 - 9$, $-2 \leq x \leq 2$.
- (i) Find the equation of the inverse function $f^{-1}(x)$. **1**
- (ii) On the same diagram, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$, showing clearly the coordinates of the endpoints and the intercepts on the axes. **3**
- (b) (i) State the domain and range of $y = \cos^{-1}\left(\frac{5x}{3}\right)$ **2**
- (ii) Hence sketch the graph of $y = \cos^{-1}\left(\frac{5x}{3}\right)$ **1**
- (c) Evaluate $\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-4x^2}}$ in exact form. **2**
- (d) Consider the function $f(x) = 2 \tan^{-1} x + \sin^{-1}(\log_e x)$ where $x \geq 0$.
Find $f'(x)$. **2**
- (e) Find the general solution of the equation $\cos x = \frac{\sqrt{3}}{2}$.
Express your answer in terms of π . **1**

Question 5 (12 Marks) START A NEW PAGE**Marks**

(a) (i) Show that $\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$. 1

- (ii) The velocity, $v \text{ ms}^{-1}$ of a particle moving in a straight line is given by $v = \sqrt{25 - x^2}$, where x is the displacement in metres from O . Show that the acceleration is $\ddot{x} = -x$. 1

- (b) When $x \text{ cm}$ from the origin, the acceleration of a particle moving in a straight line is given by:

$$\frac{d^2x}{dt^2} = \frac{-5}{(x+2)^3}$$

It has an initial velocity of 2 cm/s at $x=0$. If the velocity is $V \text{ cm/s}$, find V in terms of x . 2

- (c) A particle is moving in simple harmonic motion in a straight line. At time t seconds, it has displacement x metres from a fixed point O in the line, velocity $\dot{x} \text{ ms}^{-1}$ given by $\dot{x} = -12 \sin(2t + \frac{\pi}{3})$ and acceleration $\ddot{x} \text{ ms}^{-2}$. Initially the particle is 5 metres to the right of O .

- (i) Show that $\ddot{x} = -4(x - 2)$. 1
- (ii) Find the period and the extremities of the motion. 2
- (iii) Find the time taken by the particle to return to its starting point for the first time. 1

- (d) A rock is hurled from the top of a 15 m cliff with an initial velocity of 26 ms^{-1} at an angle of projection equal to $\tan^{-1} \left(\frac{5}{12} \right)$ above the horizontal.

The cliff overlooks a flat paddock.

The equations of motion of the stone are $\ddot{x} = 0$ and $\ddot{y} = -10$

- (i) Taking the origin as the base of the cliff, show the components of the rock's displacement are $x = 24t$ and $y = -5t^2 + 10t + 15$. 2
- (ii) Calculate the time until impact with the paddock, and the distance of the point of impact from the base of the cliff. 2

Question 6 (12 Marks) START A NEW PAGE**Marks**

- (a) Prove by Mathematical Induction that,
 $(n)^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all positive whole numbers n . **3**
- (b) Three consecutive coefficients in the expansion of $(1+x)^n$ are in the ratio 6:3:1.
(i) Find the value of n . **4**
(ii) State which terms have their coefficients in the ratio 6:3:1. **1**
- (c) Let n and m be positive integers with $m \leq n \leq 2m+1$.
(i) Show that $(1+x)^{n-m} \left(1+\frac{1}{x}\right)^m = \frac{\left(1+\frac{1}{x}\right)^n}{x^{m-n}}$ **1**
(ii) By applying the binomial theorem to part (i) and equating the coefficient of x , find a simpler expression for
 ${}^{n-m}C_1 {}^mC_0 + {}^{n-m}C_2 {}^mC_1 + {}^{n-m}C_3 {}^mC_2 + \dots + {}^{n-m}C_{n-m} {}^mC_{n-m-1}$ **3**

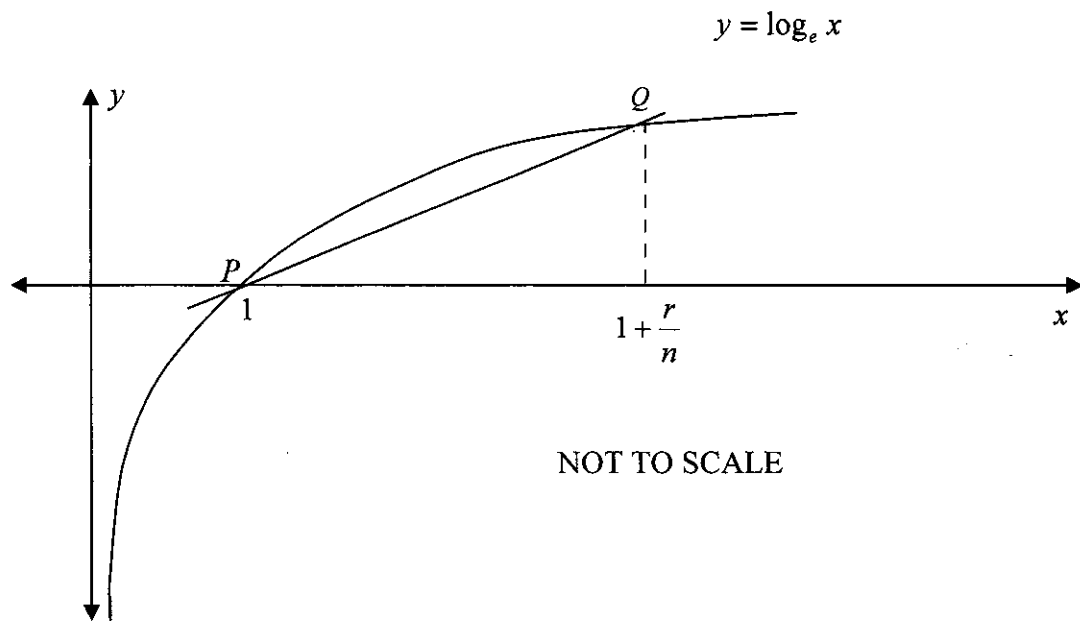
Question 7 (12 Marks) START A NEW PAGE

- (a) The equation $f(x) = \sqrt{x} + \sqrt{x+1} + \sqrt{x+2} - 5$ has a root α between $x = 1.5$ and $x = 2$.
Find the interval in which α lies by applying halving the interval twice. **1**
- (b) A spherical bubble is expanding so that its volume is increasing at the constant rate of $10 \text{ mm}^3 / \text{s}$. What is the rate of increase of the radius when the surface area is 500 mm^2 ? **2**
- (c) After t hours, the number of individuals in a population is given by $N = 500 - 400e^{-0.1t}$.
(i) Sketch the graph of N as a function of t , showing clearly the initial population size and the limiting population size. **2**
(ii) Show that $\frac{dN}{dt} = 0.1(500 - N)$. **1**
(iii) Find the population size for which the rate of growth of the population is half the initial rate of growth. **1**

Question 7 continues on the page 7.....

Question 7 continued.....

- (d) The diagram below shows the graph of $y = \log_e x$ and the secant joining points P and Q on the curve. P is at $x = 1$ and Q is at $x = 1 + \frac{r}{n}$.



- (i) Show that the gradient of the secant is $\frac{1}{r} \log_e \left(1 + \frac{r}{n} \right)^n$. **1**
- (ii) Use $\frac{d}{dx} \log_e x = \frac{1}{x}$ to show that $\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n} \right)^n = e^r$. **3**
- (iii) Use part (ii) to determine an expression for the effective annual rate of interest when an annual rate of 6% p.a. is compounded continually, that is, compounded an infinite number of times per year. **1**

END OF EXAMINATION

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONS

1. a) $P(-2, 3)$ $Q(6, -1)$

$$R = \left(\frac{3 \cdot 2}{3+2}, \frac{2 \times 3 + 6 \times (-1)}{3+2} \right) \quad \textcircled{1}$$

$$= \left(\frac{14}{5}, \frac{2}{5} \right) \quad \textcircled{1}$$

b) $r = \frac{e}{e+1}$, $a = \frac{e}{e+1}$

Limiting Sum

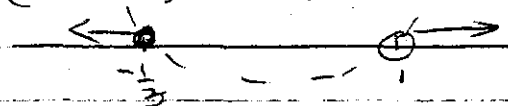
$$= \frac{e}{1+e} \times \frac{e+1}{e+1-e} \quad \textcircled{1}$$

$$= \frac{e(e+1)}{(1+e)} \quad \textcircled{1}$$

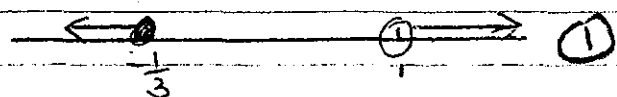
$$= e \quad \textcircled{1}$$

c) $(1-x)^2 \times \frac{4}{1-x} \leq 3(1-x)^2, x \neq 1$

$$\left. \begin{aligned} 4(1-x) &\leq 3(1-2x+x^2) \\ 4-4x &\leq 3-6x+3x^2 \\ 3x^2-2x-1 &\geq 0 \\ (3x+1)(x-1) &\geq 0 \end{aligned} \right\} \quad \textcircled{1}$$



$$x \leq -\frac{1}{3} \text{ or } x \geq 1 \quad \textcircled{1}$$



d) $\alpha + \beta + \gamma = -2$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 3$$

$$\alpha\beta\gamma = -6$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{3}{-6}$$

$$\textcircled{1} = -\frac{1}{2} \quad \textcircled{1}$$

e) $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{2}$$

$$= \frac{3}{2} \quad \textcircled{1}$$

f) $y = \frac{1}{\sqrt{3}}x + \frac{1}{\sqrt{3}}$

$$y = x - 4$$

$$m_1 = \frac{1}{\sqrt{3}}, m_2 = 1$$

$$\tan \theta = \left| \frac{\frac{1}{\sqrt{3}} - 1}{1 + \frac{1}{\sqrt{3}}} \right| \quad \textcircled{1}$$

$$\theta = 15^\circ \quad \textcircled{1}$$

82 (a) (i) $m = \frac{p+q}{2}$

Equation of PQ

$$y - aq^2 = \frac{p+q}{2} (x - 2aq)$$

$$2y - 2aq^2 = (p+q)x - 2apq - 2aq^2$$

$$2y = (p+q)x - 2apq \quad \text{given}$$

(ii) $x = 0, y = -2a$

$$-4a = (p+q) \times 0 - 2apq$$

$$2apq = 4a$$

$$pq = \frac{4a}{2a}$$

$$\therefore pq = 2$$

(iii) $K(-apq(p+q), a(p^2+q^2+pq+2))$

Since $pq = 2$

$$K(-2a(p+q), a(p^2+q^2+4))$$

$$x = -2a(p+q), \quad y = a(p^2+q^2+4)$$

$$= a[(p+q)^2 - 2pq + 4]$$

$$x^2 = 4a^2(p+q)^2$$

$$y = a(p+q)^2$$

$$\therefore x^2 = 4ay$$

b) (i) given $\angle DEC = \angle ECD$

$$\angle ECD = \angle EBD \quad (\angle's \text{ at circumference equal}$$

subtended by same arc ED)

$$\therefore \angle CED = \angle EBD$$

Question 2

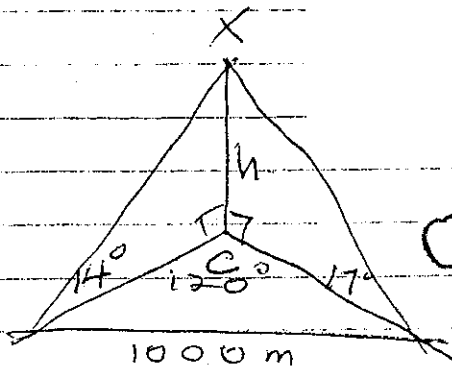
- 1) $\angle EDA = \angle ABE$ ($\angle$'s at circumference subtended by arc AE equal) ①
 $\angle GFD = \angle CED + \angle EDA$ (exterior $\angle$ of $\triangle EFD$ equal
 sum of opposite interior $\angle$'s) ①

$$\therefore \angle ABG = \angle ABE + \angle EBD \text{ (adjacent } \angle\text{'s)} \\ = \angle EDA + \angle ECD \text{ (from part (i))} \\ = \angle GFD \quad \text{①}$$

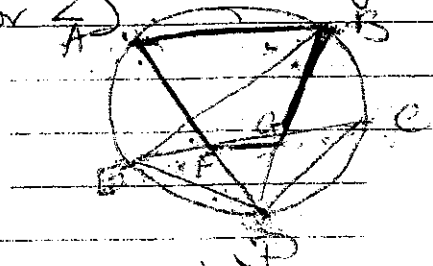
$\angle GFD$ is exterior $\angle$ of quadrilateral ABGF
 $\angle ABG$ is opposite interior $\angle$ to $\angle GFD$ in
 quadrilateral ABGF

$\therefore$ ABGF is a cyclic quadrilateral (exterior $\angle$ of cyclic
 quadrilateral equal opposite interior $\angle$)

2) (i)



① (must have information)



(ii) $\tan 14^\circ = \frac{h}{AC}$

$$AC = h \cot 14^\circ$$

$$\tan 17^\circ = \frac{h}{BC}$$

$$BC = h \cot 17^\circ$$

$$1000^2 = h^2 \cot^2 14^\circ + h^2 \cot^2 17^\circ - 2 \times h^2 \cot 14^\circ \cot 17^\circ \cos 120^\circ \quad \text{①}$$

$$= h^2 (\cot^2 14^\circ + \cot^2 17^\circ - 2 \cot 14^\circ \cot 17^\circ \cos 120^\circ)$$

$$h^2 = \frac{1000^2}{\cot^2 14^\circ + \cot^2 17^\circ + \cot 14^\circ \cot 17^\circ} \quad \text{①}$$

$$= 25060.44965$$

$$h = 158.3049262$$

$$= 158 \text{ m (nearest m)}$$

Question 3

a) (i) Vertical asymptotes

$$1-x^2=0$$

$$(1-x)(1+x)=0$$

$\therefore x=1, x=-1$ are asymptotes ① for both
Other asymptote

$$y = \lim_{x \rightarrow \infty} \frac{3x}{1-x^2}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3}{x}}{\frac{1}{x^2}-1}$$

$$= 0$$

$\therefore y=0$ is an asymptote ①

$$(ii) h'(x) = \frac{3(1-x^2) + 3x \times 2x}{(1-x^2)^2}$$

$$= \frac{3-3x^2+6x^2}{(1-x^2)^2}$$

$$= \frac{3+3x^2}{(1-x^2)^2} \quad ①$$

Since $3+3x^2 > 0$ for all values of x

$h'(x) \neq 0$ for all values of x

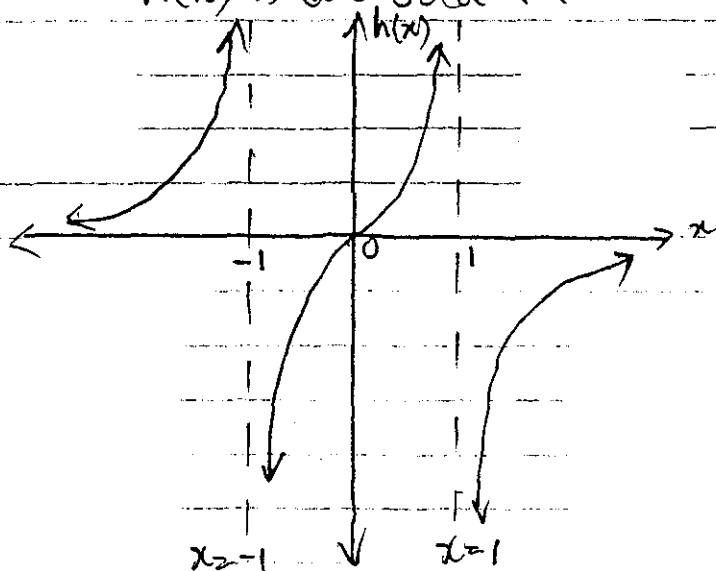
$\therefore h(x)$ does not have any turning points.

} ①

$$(iii) h(-x) = \frac{-3x}{1-x^2}$$

$$= -h(x)$$

$\therefore h(x)$ is an odd fn



Show asymptote.

Shape ①

& intercepts ①

(5)

Question 3

$$b) P(x) = k(x-1)(x+1)(x-2)$$

$$= 0$$

$$P(3) = 16$$

$$\therefore 16 = k(3-1)(3+1)(3-2)$$

$$\therefore k = 2$$

(1)

$$P(x) = 2(x-1)(x+1)(x-2)$$

$$= 2(x^2-1)(x-2)$$

$$= 2(x^3 - x - 2x^2 + 2)$$

$$= 2x^3 - 4x^2 - 2x + 4$$

(1)

$$c) V = \pi \int_0^{\pi} \sin^2 x \, dx$$

$$= \frac{\pi}{2} \int_0^{\pi} (1 - \cos 2x) \, dx$$

(1)

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi}$$

$$= \frac{\pi}{2} \left[\left(\pi - \frac{1}{2} \sin 2\pi \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi^2}{2} \text{ units}^2$$

(1)

Alternative method

Roots are 1, -1, 2

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$1 - 1 + 2 = -\frac{b}{a}$$

$$\therefore b = -2a$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$1 \times -1 + 1 \times 2 + -1 \times 2 = \frac{c}{a}$$

$$-1 + 2 - 2 = \frac{c}{a}$$

$$\therefore c = -a$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$1 \times -1 \times 2 = -\frac{d}{a}$$

$$\therefore d = 2a$$

$$P(3) = 16$$

$$a(3^3) + b(3^2) + 3c + d = 16$$

$$27a + 9b + 3c + d = 16$$

$$27a - 18a - 3a + 2a = 16$$

$$8a = 16$$

$$\therefore a = 2$$

$$b = -4, c = -2, d = 4$$

(1)

$$P(x) = 2x^3 - 4x^2 - 2x + 4$$

(1)

$$d) u = x^4, du = 4x^3 dx$$

$$\int \frac{x^3}{1+x^8} dx = \frac{1}{4} \int \frac{4x^3}{1+(x^4)^2} dx$$

$$= \frac{1}{4} \int \frac{du}{1+u^2}$$

(1)

$$= \frac{1}{4} \tan^{-1} u + c$$

$$= \frac{1}{4} \tan^{-1}(x^4) + c$$

(1)

Question 4

2) (i) $f(x) = (x+2)^2 - 9$, $-2 \leq x \leq 2$
 $0 \leq x+2 \leq 4$

$$(x+2)^2 - 9 = y$$

$$(x+2)^2 = y+9$$

$$x+2 = \sqrt{y+9}$$

$$x = -2 + \sqrt{y+9}, -9 \leq y \leq 7$$

$$\therefore f^{-1}(x) = -2 + \sqrt{x+9}, -9 \leq x \leq 7 \text{ (1)}$$

When $x+2 = 0$

$$\sqrt{y+9} = 0$$

$$y = -9$$

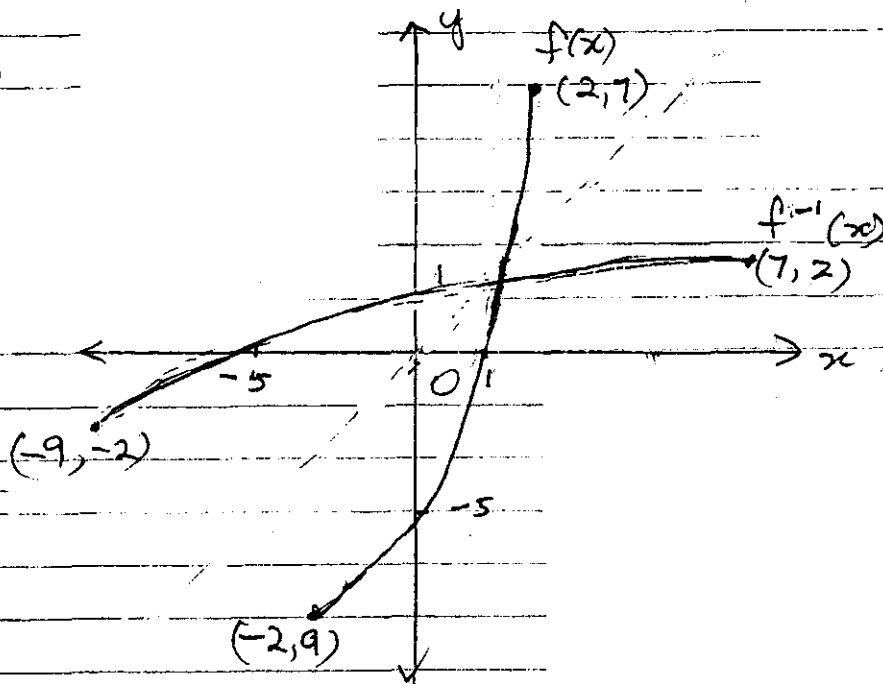
When $x+2 = 4$

$$\sqrt{y+9} = 4$$

$$y+9 = 16$$

$$y = 7$$

(ii)



Endpoints
+ intercepts (1)

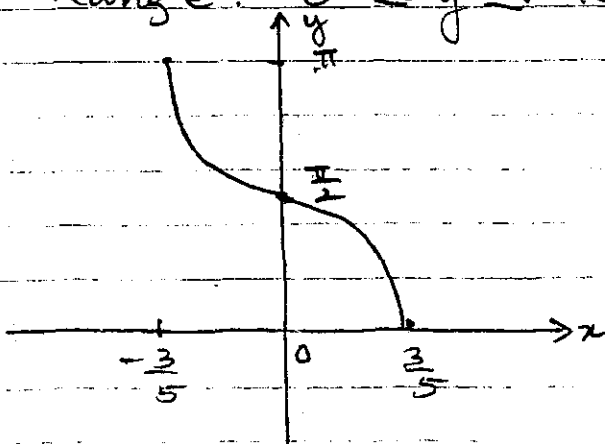
shape (1)

symmetry about $y=x$ (1)

(b) (i) $-1 \leq \frac{5x}{3} \leq 1$

Domain $-\frac{3}{5} \leq x \leq \frac{3}{5}$ (1)

Range: $0 \leq y \leq \pi$ (1)



(1) must show
domain + range

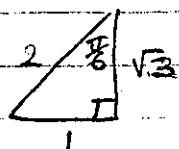
Question 4

⑦

$$\begin{aligned}
 c) \int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-4x^2}} &= \int_0^{\frac{3}{2}} \frac{dx}{\sqrt{4(\frac{9}{4}-x^2)}} \\
 &= \frac{1}{2} \int_0^{\frac{3}{2}} \frac{dx}{\sqrt{(\frac{3}{2})^2 - x^2}} \quad \text{①} \\
 &= \frac{1}{2} \left[\sin^{-1} \frac{2x}{3} \right]_0^{\frac{3}{2}} \\
 &= \frac{1}{2} [\sin^{-1} 1 - \sin^{-1} 0] \\
 &= \frac{1}{2} \times \frac{\pi}{2} \\
 &= \frac{\pi}{4} \quad \text{①}
 \end{aligned}$$

$$d) f(x) = 2 \tan^{-1} x + \sin^{-1}(\log_e x)$$

$$\begin{aligned}
 f'(x) &= \frac{2}{1+x^2} + \frac{\frac{1}{x}}{\sqrt{1-(\log_e x)^2}} \\
 &= \frac{2}{1+x^2} + \frac{1}{x \sqrt{1-(\log_e x)^2}} \quad \text{①}
 \end{aligned}$$

$$e) \cos x = \frac{\sqrt{3}}{2}$$


$$x = 2\pi n \pm \frac{\pi}{6} \quad \text{①}$$

Question 5

$$\begin{aligned}
 \text{a) (i) } \ddot{x} &= \frac{d^2x}{dt^2} \\
 &= \frac{dv}{dt} \\
 &= \frac{dv}{dx} \times \frac{dx}{dt} \\
 &= v \cdot \frac{dv}{dx} \\
 &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx} \\
 &= \frac{d}{dx} \left(\frac{v^2}{2} \right) \text{ given}
 \end{aligned}$$

①

$$\text{ii) } v = \sqrt{25 - x^2}$$

$$\begin{aligned}
 \frac{dv}{dx} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dx} \left(\frac{1}{2} (\sqrt{25 - x^2})^2 \right) \\
 &= \frac{1}{dx} \left[\frac{25 - x^2}{2} \right] \\
 &= \frac{d}{dx} \left[\frac{25}{2} - \frac{x^2}{2} \right] \\
 &= -x
 \end{aligned}$$

①

$$\therefore \ddot{x} = -x \text{ given}$$

$$\text{b) } \frac{d^2x}{dt^2} = \frac{-5}{(x+2)^3}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -5(x+2)^{-3}$$

$$\frac{1}{2} v^2 = \frac{-5}{-2(x+2)^2} + c \quad \text{①}$$

$$v = 2, x = 0$$

$$2 = \frac{5}{8} + c \quad \therefore c = \frac{11}{8}$$

$$\frac{1}{2} v^2 = \frac{5}{2(x+2)^2} + \frac{11}{8}$$

$$v^2 = \frac{5}{(x+2)^2} + \frac{11}{4} = \frac{20 + 11(x+2)^2}{4(x+2)^2}$$

①

$$v = \frac{\sqrt{20 + 11(x+2)^2}}{2(x+2)}$$

since $v > 0$ when $x = 2$

(9)

Question 5

$$1) \ddot{x} = -12 \sin(2t + \frac{\pi}{3})$$

$$x = 6 \cos(2t + \frac{\pi}{3}) + c$$

$$\text{When } t=0, x=5$$

$$5 = 6 \cos \frac{\pi}{3} + c$$

$$5 = 3 + c$$

$$\therefore c = 2$$

$$\therefore x = 2 + 6 \cos(2t + \frac{\pi}{3})$$

$$6 \cos(2t + \frac{\pi}{3}) = x - 2$$

$$\ddot{x} = -12 \times 2 \cos(2t + \frac{\pi}{3})$$

$$= -24 \cos(2t + \frac{\pi}{3})$$

$$= -4 \times 6 \cos(2t + \frac{\pi}{3})$$

$$= -4(x - 2) \text{ given}$$

①

$$\text{ii) Extremities: } x = 2 + 6 \times \pm 1$$

$$-4 \leq x \leq 8$$

①

$$\text{Period: } \frac{2\pi}{2} = \pi \text{ seconds}$$

①

$$\text{ii) When } t=0, x=5$$

$$5 = 2 + 6 \cos(2t + \frac{\pi}{3})$$

$$3 = 6 \cos(2t + \frac{\pi}{3})$$

$$\cos(2t + \frac{\pi}{3}) = \frac{1}{2}$$

$$2t + \frac{\pi}{3} = \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \dots$$

$$2t = 0, \frac{4\pi}{3}, \dots$$

$$t = 0, \frac{2\pi}{3}$$

①

First return after $\frac{2\pi}{3}$ seconds.

Question 5

g) If $\tan \theta = \frac{5}{12}$
 $\cos \theta = \frac{12}{13}$
 $\sin \theta = \frac{5}{13}$

(i) $\ddot{x} = 0, \dot{x} = \int \ddot{x} dt$
 $= C_1$

$\dot{x} = v \cos \theta$
 $= 26 \times \frac{12}{13}$
 $= 24 \text{ when } t=0, C_1 = 0$

$x = \int \dot{x} dt$
 $= \int 24 dt$

$= 24t + C_2$

$x=0, t=0 \therefore C_2 = 0$

$x = 24t$ Given

$\ddot{y} = -10$

$\dot{y} = \int \ddot{y} dt$
 $= -10t + C_3$

$\dot{y} = v \sin \theta$
 $= 26 \times \frac{5}{13}$
 $= 10$

When $t=0, \dot{y}=10, C_3 = 10$

$\dot{y} = -10t + 10$

$y = \int (-10t + 10) dt$
 $= -5t^2 + 10t + C_4$

When $t=0, y=15, C_4 = 15$

$y = -5t^2 + 10t + 15$ Given

(ii) Impact when $y=0$

$-5t^2 + 10t + 15 = 0$

$t^2 - 2t - 3 = 0$

$(t+1)(t-3) = 0$

$\therefore t = -1, 3 \text{ But } t \geq 0$

$\therefore t = 3$

When $t=3$

$x = 24 \times 3$

$= 72 \text{ m}$

(11)

Q6

a) when $n=1$, $n^3 + (n+1)^3 + (n+2)^3 = 1^3 + 2^3 + 3^3$
 $= 1 + 8 + 27$
 $= 36$
 $= 4 \times 9$

$\therefore$ It is true for $n=1$.

Assume the statement is true for $n=k$, where k is a positive integer.

i.e. $k^3 + (k+1)^3 + (k+2)^3 = 9m$ for some positive integer m
 $(k+1)^3 + (k+2)^3 = 9m - k^3$ (1)

then $(k+1)^3 + (k+2)^3 + (k+3)^3$
 $= 9m - k^3 + (k+3)^3$ by assumption (1) ①
 $= 9m + 3[(k+3)^2 + k(k+3) + k^2]$
 $= 9m + 3[k^2 + 6k + 9 + k^2 + 3k + k^2]$
 $= 9m + 3(3k^2 + 9k + 9)$
 $= 9m + 9(k^2 + 3k + 3)$
 $= 9(m + k^2 + 3k + 3)$ ①

which is divisible by 9.

It will be true for $n=k+1$ if it is true for $n=k$.
 Since it is proved true for $n=1$, $\therefore$ it will be true
 for $n=2, 3, 4, \dots$, i.e. true for all positive integers n .

(b) $(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$

(i) Let the 3 consecutive terms be $\binom{n}{k-1} x^{k-1}$, $\binom{n}{k} x^k$ & $\binom{n}{k+1} x^{k+1}$

$\therefore \binom{n}{k-1} : \binom{n}{k} : \binom{n}{k+1} = 6 : 3 : 1$

$\frac{n!}{(n-k+1)!(k-1)!} : \frac{n!}{(n-k)!k!} : \frac{n!}{(n-k-1)!(k+1)!} = 6 : 3 : 1$ ①

$$\frac{k}{n-k+1} = \frac{6}{3} \rightarrow \textcircled{1}$$

$$\frac{k}{n-k+1} = 2$$

$$k = 2n - 2k + 2$$

$$3k = 2n + 2 \quad (1)$$

and $\frac{k+1}{n-k} = \frac{3}{1} \rightarrow \textcircled{1}$

$$k+1 = 3n - 3k$$

$$4k = 3n - 1 \quad (2)$$

$$(1) \times 4 \quad 12k = 8n + 8 \quad (3)$$

$$(2) \times 3 \quad 12k = 9n - 3 \quad (4)$$

$$(4) - (3) \quad 0 = n - 11$$

$$n = 11 \rightarrow \textcircled{1}$$

(ii) Put $n=11$ into (1)

$$3k = 22 + 2$$

$$k = 8$$

$\therefore$ The terms are the 8th, 9th and 10th terms. $\textcircled{1}$

(c) (i) $(1+x)^{n-m} \cdot \left(1+\frac{1}{x}\right)^m = (1+x)^{n-m} \left(\frac{1+x}{x}\right)^m$

$$= \frac{(1+x)^n}{x^m}$$

$$= \left(\frac{1+x}{x}\right)^n \cdot x^n \cdot \frac{1}{x^m}$$

$$= \left(1+\frac{1}{x}\right)^n \cdot \frac{x^n}{x^m}$$

$$= \frac{\left(1+\frac{1}{x}\right)^n}{x^{m-n}} \quad \text{Given} \quad \textcircled{1}$$

(13)

$$(ii) \quad (1+x)^{n-m} \left(1+\frac{1}{x}\right)^m = \sum_{r=0}^{n-m} \binom{n-m}{r} x^r \cdot \sum_{j=0}^m \binom{m}{j} \frac{1}{x^j}$$

$$\therefore \text{coeff. of } x \text{ in the expansion} \\ = \binom{n-m}{1} \binom{m}{0} + \binom{n-m}{2} \binom{m}{1} + \binom{n-m}{3} \binom{m}{2} + \dots + \binom{n-m}{n-m} \binom{m}{n-m-1} \quad (1)$$

$$\frac{1}{x^{n-n}} \left(1+\frac{1}{x}\right)^n = \frac{1}{x^{n-n}} \sum_{k=0}^n \binom{n}{k} \frac{1}{x^k}$$

$$= \sum_{k=0}^n \binom{n}{k} x^{n-m-k}$$

$$\therefore \text{coeff. of } x \text{ in the expansion} = \binom{n}{n-m-1} \quad (1)$$

$$\text{Equating coeff of } x \text{ on both sides of } (1+x)^{n-m} \left(1+\frac{1}{x}\right)^m = \frac{\left(1+\frac{1}{x}\right)^n}{x^{m-n}}$$

$$\binom{n-m}{1} \binom{m}{0} + \binom{n-m}{2} \binom{m}{1} + \binom{n-m}{3} \binom{m}{2} + \dots + \binom{n-m}{n-m} \binom{m}{n-m-1}$$

$$= \binom{n}{n-m-1}$$

$$i.e. \quad {}^{n-m}C_1 \cdot {}^mC_0 + {}^{n-m}C_2 \cdot {}^mC_1 + {}^{n-m}C_3 \cdot {}^mC_2 + \dots + {}^{n-m}C_{n-m} \cdot {}^mC_{n-m-1} = {}^nC_{n-m-1}$$

(1)

Question 7

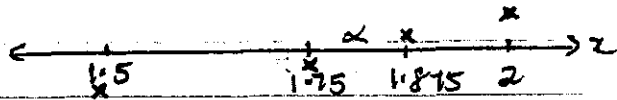
$$(a) f(x) = \sqrt{x} + \sqrt{x+1} + \sqrt{x+2} - 5$$

$$f(1.5) = -0.323297605$$

$$f(2) = 0.146254369$$

$$\text{Half the interval } x = \frac{1.5+2}{2} = 1.75$$

$$f(1.75) = -0.082320276$$



$$\text{Half interval } x = \frac{1.75+2}{2} = 1.875$$

$$f(1.875) = 0.033390859$$

$$\therefore 1.75 < x < 1.875$$

①

b) V = volume of sphere, r = radius

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \times \frac{dV}{dt}$$

①

$$\frac{dV}{dt} = 10$$

$$\therefore \frac{dr}{dt} = \frac{10}{4\pi r^2}$$

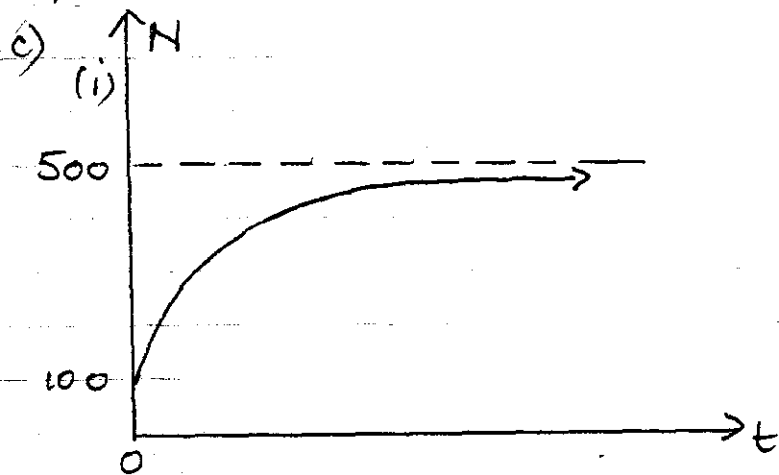
When surface area = 500 mm^2

$$4\pi r^2 = 500$$

$$\therefore \frac{dr}{dt} = \frac{10}{500}$$

$$= 0.02 \text{ mm/s.}$$

①



① Shape

① asymptote & intercept

(ii) $N = 500 - 400e^{-0.1t}$
 $\frac{dN}{dt} = 0.1 \times 400e^{-0.1t}$ ①
 $= 0.1(500 - N)$

(iii) Initial rate of growth is
 $0.1(500 - 100)$
 $= 0.1 \times 400$

When population is half
 $= 0.1 \times 200$

$\therefore 0.1(500 - N) = 0.1 \times 200$

$500 - N = 200$

$N = 300$ ①

c) (i) At Q, $y = \log_e(1 + \frac{r}{n})$

Gradient of PQ = $\frac{\log_e(1 + \frac{r}{n}) - 0}{(1 + \frac{r}{n}) - 1}$

$= \frac{n}{r} \log_e(1 + \frac{r}{n})$

$= \frac{1}{r} \log_e(1 + \frac{r}{n})^n$ } ①

Question 7

d) ii) $\frac{d}{dx}(\log_e x) = \frac{1}{x}$

at $x=1$, gradient at $P = 1$

as $\rightarrow \infty$, PQ $\rightarrow$ tangent at P

} (1)

$\therefore \lim_{n \rightarrow \infty} \frac{1}{r} \log_e \left(1 + \frac{r}{n}\right)^n = 1$

(1)

$\lim_{n \rightarrow \infty} \log_e \left(1 + \frac{r}{n}\right)^n = r$

(1)

$\lim_{n \rightarrow \infty} e^{\log_e \left(1 + \frac{r}{n}\right)^n} = e^r$

$\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n = e^r$

iii) If compound interest is paid n times per year at 6% p.a., then

$A = P \left(1 + \frac{0.06}{n}\right)^n$

as $n \rightarrow \infty$, interest is compounded continually

$A = \lim_{n \rightarrow \infty} P \left(1 + \frac{0.06}{n}\right)^n$

$A = P \times e^{0.06}$

$1+r = e^{0.06}$

$\therefore$ effective rate when compounded continuously is $r = e^{0.06} - 1$

(1)